

6. You can expect to see wearable electronics using bioengineering, nanotechnology and 3D scanning techniques. These technologies will enable your clothes to control air flow and open invisible shutters when the humidity goes up. Similarly, interconnection technologies using conductive fabrics will enable you to adjust the colour and texture of your drapes by simply pressing a button!

7. A sugarcube-sized electric generator that feeds on vibration could power a number of wireless sensors or even medical implants.

8. Virtual travelling and picnics will be possible with the advancements in electronics.

9. Technological advancements in electronics may change the face of gifts. For example, a ring which conceals a high precision computer with Global Positioning System (GPS) that constantly monitors its location.

10. A software may enable you to block credit cards at the entrances of exclusive shops.

11. Turbo liver, surgically connected to the circulatory system of the body, will limit the growth of beer belly.

12. Advancements in transparent electronics will help in space management by allowing electronic devices to be stacked in small clear spaces. It will also enable development of clear computer monitors and televisions that can be embedded inside glass or transparent plastic (or flexible materials). Recently, Korea Advanced Institute of Science and Technology has developed a transparent computer chip using the transparent resistive random access memory (1T1R) technology. Besides being transparent, it functions as complementary metal-oxide semiconductor (CMOS) memory chip, providing non-volatile memory.

13. Touch interface electronics is expected to find application in majority of displays, including walls and desks.

14. All electronic devices will be service-connected, which will relieve

users from bridging content and devices.

15. Future automobiles are expected to be equipped with GPS devices and electrically-driven propulsion systems with no emission other than water.

Emerging technologies

Carbon nanostructures. Electrical and optical properties of nanoscale carbon materials (including carbon nanotubes) promise exciting applications of carbon nanostructures in future electronic products. Studies regarding properties of nanotubes and graphene aim at developing a technology that will make devices more compact, fast and energy efficient.

Research in nano particles is making fast advancements. Now storage of information on a chip has become much cheaper than on paper. The optoelectronic studies aim at integrating this technology with an optical technology based on the same materials. The research will aid in the development of future high-speed electronics, communications systems, and sensors for diverse applications. Industries ranging from automobile, aviation, space and energy conversion/conservation to bionanotechnology and medicine are likely to benefit from the research.

Spintronics. Even as conventional electronic products are based on the charge of the electrons, attempts to use their spin has given rise to a new technology known as Spintronics. It is an acronym for spin transport electronics, and will have a significant impact on the future of electronics.

Conventional electronics exploits electrons in a conducting material, but for the same number of electrons, if their spin is exploited, the same unit becomes multipurpose. Control on two spins of electrons offer a number of possibilities in terms of permutations and combinations.

Plastic (flexible) electronics. Polymers that conduct electricity without the assistance of fillers are quickly emerging as the basis for future de-

DigiPower
OKAYA®

Sealed Maintenance Free VRLA Battery

The Power to Perform



**Sealed Maintenance Free
VRLA Batteries**

RANGE : 7.2Ah to 200Ah

Break Free Life

A clutch of design features ensures that OKAYA SMF / VRLA batteries perform predictably and reliably every time.

- High Power Density
- Excellent Recovery From Deep Discharge
- Reliable Construction
- Maintenance Free Operation
- Higher Reliability & Quality

OKAYA POWER LTD.

(SMF / VRLA BATTERY INDIA)
A-42, Sector-63, Noida (UP) INDIA
Tel.: 0120-4640800 Fax: 0120-4640827
Email: vrla@okayapower.com

BULK BUYER CONTACT
DISTRIBUTORS ENQUIRY INVITED
FOR UNREPRESENTED AREAS